1. Payment Processing, Refund SLAs & Automated Reversal Architecture

Executive Summary: *NovaPay operates a high-throughput distributed payment switch processing over 45,000 transactions per second (TPS). This section defines the deterministic Service Level Agreements (SLAs) for standard customer refunds, instant payout rails, merchant settlement cycles, and automated reconciliation reversals for uncaptured debits.*

1.1 Standard & Instant Customer Refund Timelines

All standard customer refunds initiated via the NovaPay Merchant Dashboard or the REST Refund API (/v2/refunds) are processed within 5 to 7 business days. Once a refund request is validated, NovaPay dispatches the ISO-8583 / ISO-20022 reversal payload to the acquiring bank switch within 24 hours. Every refund generates an immutable Acquirer Reference Number (ARN) within 48 hours. For merchants enabling Instant Refunds over UPI or IMPS rails, funds settle into the end-customer bank account within 15 minutes (p95 latency: 42 seconds) for a flat operational fee of $0.25 USD (or INR 20) per instant refund transaction.

1.2 Merchant Settlement Cycles (T+N Clearing Schedule)

Merchant clearing cycles are assigned dynamically based on business underwriting risk profiles. Tier-1 Enterprise merchants receive automated daily settlements on a T+1 business day cycle at 10:00 AM IST/UTC. Standard SMB merchants operate on a T+2 business day settlement cycle. High-risk cross-border, gaming, or travel merchants are placed on a T+5 rolling reserve cycle, wherein 10% of daily gross merchandise value (GMV) is held in an escrow reserve account for 90 calendar days to cover potential chargeback liabilities.

1.3 Pending Capture & Failed Transaction Auto-Reversals

If a customer's issuing bank account is debited during checkout but the payment state remains PENDING_CAPTURE or AUTHORIZED_UNSETTLED for more than 45 minutes without an explicit merchant capture call, the NovaPay reconciliation daemon automatically triggers an void-and-reverse workflow. The full debited principal is credited back to the customer's source account within 3 to 5 business days without requiring manual customer support intervention. Partial captures are supported up to 100% of the authorized amount within a 7-day authorization validity window.

Table 1.1: NovaPay Refund & Settlement SLA Matrix

Workflow / Rail	Target SLA	Fee / Reserve	Identifier / State Code
Standard Card / Netbanking Refund	5 - 7 Business Days	$0.00 (Free)	REFUND_STANDARD_ARN
Instant Refund (UPI / IMPS / FastRails)	Within 15 Minutes	$0.25 / INR 20 flat	REFUND_INSTANT_FAST
Auto-Reversal (Uncaptured > 45 mins)	3 - 5 Business Days	$0.00 (Automated)	AUTO_REV_TIMEOUT_45M
Tier-1 Enterprise Settlement	T+1 Business Day	0% Rolling Reserve	SETTLE_TIER1_T1
Standard SMB Settlement	T+2 Business Days	0% Rolling Reserve	SETTLE_SMB_T2
High-Risk Rolling Reserve Settlement	T+5 Business Days	10% Held for 90 Days	SETTLE_RESERVE_T5

Policy Reference FIN-101: Merchants must idempotently pass a unique X-Idempotency-Key header (UUIDv4) on all POST /v2/refunds calls. Duplicate keys within 24 hours return cached HTTP 200 responses without double-refunding.

2. API Gateway Rate Limiting, Token-Bucket Quotas & Traffic Shaping

Executive Summary: *To protect core ledger microservices and upstream banking switches from noisy-neighbor traffic spikes and credential-stuffing DDoS attacks, the NovaPay Edge Gateway enforces distributed Redis-backed Token-Bucket rate limiting across all REST, GraphQL, and SSE streaming endpoints.*

2.1 Distributed Redis Token-Bucket Quotas by Account Tier

Rate limits are evaluated atomically in under 0.4 milliseconds using pre-compiled Redis Lua scripts keyed by Merchant API Key and client IP subnet (/24). Free Sandbox accounts are restricted to 60 requests per minute (RPM) with a burst capacity of 15 requests. Production Growth plans are provisioned with 600 RPM and a burst capacity of 100 concurrent requests. Enterprise Dedicated plans receive 3,000 RPM with a burst capacity of 500 requests, expandable to 10,000 RPM during pre-scheduled flash sales upon 48 hours prior notice to the Infrastructure SRE team.

2.2 HTTP 429 Too Many Requests & Full-Jitter Exponential Backoff

When a client exhausts its token bucket, the API Gateway immediately rejects the request with HTTP 429 Too Many Requests and injects three mandatory response headers: X-RateLimit-Limit, X-RateLimit-Remaining, and Retry-After (specifying cooldown duration in integer seconds). Client SDKs must implement exponential backoff with full jitter starting at a base delay of 500 milliseconds up to a maximum cap of 32 seconds: sleep_ms = random_uniform(0, min(32000, 500 * 2^attempt)).

2.3 Bulk Batch Payout & Reporting Endpoint Quotas

High-compute asynchronous endpoints operate on isolated token buckets. The Bulk Payout endpoint (/v2/payouts/batch) accepts up to 500 transfer instructions per single JSON payload and is rate-limited to 20 batch calls per minute (equivalent to 10,000 payouts/minute). Reconciliation CSV/PDF export endpoints (/v2/reports/export) are strictly limited to 5 requests per minute per merchant tenant.

Table 2.1: API Gateway Rate Limit Tiers & Backoff Parameters

Account Tier / Endpoint	Sustained Quota (RPM)	Burst Capacity	Throttle Status & Header
Free Sandbox Tier	60 RPM (1 req/sec)	15 Requests	HTTP 429 + Retry-After
Production Growth Tier	600 RPM (10 req/sec)	100 Requests	HTTP 429 + Retry-After
Enterprise Dedicated Tier	3,000 RPM (50 req/sec)	500 Requests	HTTP 429 + Retry-After
Enterprise Flash-Sale Override	10,000 RPM (166 req/sec)	1,500 Requests	Requires 48h SRE Pre-Warm
Bulk Batch Payouts (/v2/payouts/batch)	20 Batch Calls / Min	500 Transfers / Call	Isolated Redis Bucket
Client Backoff Policy	Base: 500ms, Cap: 32s	Full Jitter Required	Max 6 Retries Recommended

Policy Reference GW-204: Spammers exceeding 5x their allocated RPM quota for more than 120 consecutive seconds trigger an automated WAF block for 15 minutes (Error Code: ERR_WAF_RATE_ABUSE).

3. Chargeback Lifecycle, Dispute Representment & Fraud Thresholds

Executive Summary: *NovaPay provides an automated Dispute Representment Engine integrated directly with Visa Verifi (CDRN) and Mastercard Ethoca networks. Merchants must adhere strictly to card network evidence submission timelines and dispute ratio thresholds to preserve processing privileges.*

3.1 Mandatory 14-Day Chargeback Response Window

When an issuing bank initiates a formal chargeback (reason codes 10.4 Fraud, 13.1 Merchandise Not Received, or 13.3 Not as Described), NovaPay emits a dispute.created webhook and freezes the disputed amount in the merchant's settlement ledger. Merchants have exactly 14 calendar days from the chargeback issuance timestamp to upload compelling evidence via the Disputes API (/v2/disputes/{id}/evidence). Acceptable evidence includes signed carrier delivery receipts, customer IP/device fingerprint logs, and 3DS2 cryptographic authentication values (CAVV/ECI). Failure to submit evidence within 14 calendar days results in an irreversible default loss.

3.2 Dispute Operational Fees & Representment Resolution Timeline

Every formal chargeback filed against a merchant incurs a non-refundable operational dispute processing fee of $15.00 USD (or INR 1,250 for domestic Indian transactions), regardless of the final outcome. Once representment evidence is submitted, the card network and issuing bank adjudicate the claim within 30 to 45 calendar days. If ruled in the merchant's favor, the frozen transaction principal is released back to the merchant in the next T+1 cycle.

3.3 High-Risk Dispute Monitoring Program (0.90% Threshold)

In alignment with Visa Dispute Monitoring Program (VDMP) rules, any merchant whose rolling 30-day dispute-to-transaction count ratio exceeds 0.90% (and at least 100 disputes per month) is automatically enrolled in the NovaPay High-Risk Monitoring Program. Enrolled merchants incur an additional $25.00 USD high-risk network penalty per dispute and must enable mandatory 3DS2 Step-Up authentication on 100% of checkout traffic.

Table 3.1: Chargeback SLAs, Fees & Network Compliance Thresholds

Dispute Metric / Stage	Threshold / Timeline	Financial Impact	System Action
Merchant Evidence Submission Window	14 Calendar Days	Principal Frozen in Ledger	Auto-Loss if Missed
Standard Dispute Processing Fee	Per Chargeback Filed	$15.00 USD / INR 1,250	Non-Refundable Operational Fee
Issuer Representment Adjudication	30 - 45 Calendar Days	Principal Restored on Win	Webhook: dispute.won / lost
Early Warning Fraud Alert	0.65% Rolling 30-Day Ratio	$0 Penalty (Warning Notice)	Triggers Risk Review Ticket
High-Risk Monitoring Threshold	0.90% Rolling 30-Day Ratio	+$25.00 USD Penalty / Dispute	Mandatory 3DS2 Enforcement
Account Termination Threshold	1.50% Rolling 30-Day Ratio	100% Settlement Hold (180d)	Immediate MID Deactivation

Policy Reference RISK-309: Pre-dispute inquiries resolved via Rapid Dispute Resolution (RDR) within 48 hours are refunded automatically and do NOT count toward the merchant's 0.90% chargeback ratio.

4. Asynchronous Webhook Delivery, HMAC-SHA256 Security & DLQ Policy

Executive Summary: *NovaPay relies on an event-driven Kafka + Rust webhook dispatcher to notify merchant backends of asynchronous state transitions (payment.captured, refund.processed, dispute.created, subscription.charged). All webhook consumers must implement idempotent processing.*

4.1 Delivery Guarantees & Strict 5,000ms Timeout SLA

NovaPay guarantees at-least-once delivery for all subscribed webhook events. When an event is dispatched to a merchant's HTTPS webhook URL, the merchant server must respond with an HTTP 2xx status code (200 OK or 202 Accepted) within a strict 5,000 millisecond (5-second) timeout window. Any response taking longer than 5,000ms, or returning HTTP 3xx redirects, HTTP 4xx, or HTTP 5xx status codes, is marked as a failed delivery attempt.

4.2 8-Stage Exponential Retry Schedule Over 24 Hours

When a webhook delivery fails or times out, the dispatcher places the event into a delayed retry queue and attempts delivery up to 8 additional times over a 24-hour period following this exact schedule: Attempt 1 at +30 seconds, Attempt 2 at +2 minutes, Attempt 3 at +10 minutes, Attempt 4 at +30 minutes, Attempt 5 at +1 hour, Attempt 6 at +3 hours, Attempt 7 at +6 hours, and Attempt 8 at +12 hours. After 8 consecutive failures, the webhook endpoint is marked CIRCUIT_DISABLED, an urgent email alert is sent to the merchant DevOps contact, and events are routed to the Dead-Letter Queue (DLQ) for 14 days.

4.3 Cryptographic HMAC-SHA256 Verification & Replay Protection

Every webhook HTTP POST includes two security headers: X-NovaPay-Timestamp (Unix epoch seconds) and X-NovaPay-Signature. Merchants must compute HMAC-SHA256(webhook_secret, timestamp + '.' + raw_request_body) and compare it against X-NovaPay-Signature using a constant-time string comparison (hmac.compare_digest). To prevent replay attacks, merchants must reject any webhook where the absolute difference between current server time and X-NovaPay-Timestamp exceeds 300 seconds (5 minutes).

Table 4.1: Webhook Delivery & Security Specification

Parameter / Stage	Specification	Enforcement Rule	Failure Outcome
Ack Timeout Window	5,000 ms (5.0 Seconds)	Must return HTTP 200 / 202	Triggers Retry Schedule
Retry Schedule (Attempts 1-4)	30s, 2m, 10m, 30m	Automated Backoff Queue	Retains Original Event ID
Retry Schedule (Attempts 5-8)	1h, 3h, 6h, 12h	24-Hour Total Retry Window	Marks Endpoint CIRCUIT_DISABLED
Dead-Letter Queue (DLQ) Retention	14 Calendar Days	Replayable via /v2/events/replay	Purged Permanently After 14d
Signature Algorithm	HMAC-SHA256	Payload: timestamp + '.' + body	Reject with HTTP 401 if Mismatch
Replay Protection Window	Max 300 Seconds (5 Mins)	Verify X-NovaPay-Timestamp	Reject Stale Payloads

Policy Reference HOOK-401: TLS 1.2 or TLS 1.3 with valid public CA certificates is mandatory for all webhook URLs. Self-signed certificates and plain HTTP endpoints are rejected in Production mode.

5. PCI-DSS v4.0 Level 1 Compliance, Tokenization & PII Guardrail Policy

Executive Summary: *NovaPay maintains PCI-DSS v4.0 Level 1 Service Provider certification and SOC2 Type II compliance. Policy SEC-402 mandates strict zero-trust handling of Primary Account Numbers (PAN), cryptographic key rotation, and pre-embedding PII redaction across all AI and log pipelines.*

5.1 Client-Side Card Vaulting & Stateless Tokenization

Raw 16-digit Primary Account Numbers (PAN) and 3/4-digit CVV security codes must never traverse merchant backend servers. The NovaPay Checkout JS SDK encrypts cardholder data directly in the browser using RSA-OAEP-2048 with SHA-256 and exchanges the ciphertext with the NovaPay PCI Vault for a single-use ephemeral token (prefix: tok_live_xxx) valid for exactly 15 minutes. For recurring billing, the ephemeral token is converted into a merchant-scoped permanent vault token (prefix: crd_vault_xxx) backed by Format-Preserving Encryption (FPE-FF1).

5.2 Mandatory Pre-Embedding & Log PII Redaction Guardrail (Policy SEC-402)

Under mandatory security directive SEC-402, no customer Personally Identifiable Information (PII) or sensitive secrets may be written to application logs, analytics warehouses, vector databases, or transmitted to external LLM APIs (such as OpenAI, Anthropic, or Groq). An inline high-speed regex and Luhn-checksum guardrail must execute in under 1 millisecond prior to embedding generation or external API egress, masking: (a) Email addresses as [REDACTED_EMAIL], (b) 10-digit Indian/International phone numbers as [REDACTED_PHONE], (c) 13-to-19 digit Credit/Debit cards as [REDACTED_CARD], (d) Indian PAN ([A-Z]{5}[0-9]{4}[A-Z]{1}) and Aadhaar numbers as [REDACTED_GOV_ID], and (e) Bearer/API secret keys (sk_live_*, np_live_*) as [REDACTED_KEY].

5.3 Cryptographic Key Rotation & Prompt Injection Defense

All Data Encryption Keys (DEKs) inside AWS KMS HSMs are rotated automatically every 90 calendar days using AES-256-GCM envelope encryption, while Master Key-Encryption-Keys (KEKs) rotate every 365 days. Additionally, all AI Gateway endpoints enforce pre-inference Prompt Injection & Jailbreak guardrails, immediately blocking inputs attempting system-prompt extraction, instruction override ('ignore previous instructions'), or SQL/shell injection.

Table 5.1: Policy SEC-402 Security & PII Redaction Specifications

Security Control / Entity	Technical Standard / Pattern	Replacement Token / Limit	Latency / Rotation SLA
Ephemeral Card Token (tok_live_)	RSA-OAEP-2048 Client Encryption	Valid for 15 Minutes Only	< 8ms Vault Exchange
Email Address Redaction	RFC-5322 Regex Guardrail	[REDACTED_EMAIL]	< 0.1ms Pre-Embedding
Credit Card / PAN Redaction	13-19 Digit + Luhn Check	[REDACTED_CARD]	< 0.1ms Pre-Embedding
Phone & Gov ID (PAN/Aadhaar)	E.164 / Indian Mobile / Tax Regex	[REDACTED_PHONE] / [REDACTED_GOV_ID]	< 0.1ms Pre-Embedding
API Secret Key Leak Prevention	Prefix Scanner (sk_live_, np_live_)	[REDACTED_SECRET_KEY]	Immediate Request Block / Mask
AWS KMS Key Rotation	AES-256-GCM Hardware HSM	DEK: 90 Days | KEK: 365 Days	Zero-Downtime Envelope Re-Key

Policy Reference SEC-402: Performing PII redaction BEFORE computing 384-dim sentence embeddings both guarantees zero PII leakage into vector stores and normalizes user-specific entities, increasing semantic cache hit rates by 12% to 18%.

6. Cross-Border FX Settlement, SWIFT Rails & Export Compliance (e-FIRC)

Executive Summary: *NovaPay Global enables merchants to accept international payments from 180+ countries with dynamic currency conversion, automated sanctions screening, and regulatory export compliance documentation for cross-border trade.*

6.1 Supported Currencies & FX Conversion Markup Fee

Merchants can display localized checkout prices in 135 presentment currencies and receive domestic settlements in 14 major settlement currencies (including USD, EUR, GBP, SGD, AED, AUD, CAD, JPY, and INR). Whenever a transaction requires currency conversion between the presentment currency and the merchant's settlement currency, NovaPay locks the wholesale mid-market Reuters/Bloomberg FX rate at authorization time and applies a flat 1.75% FX currency conversion fee. The locked FX rate is guaranteed for 24 hours to shield merchants from intraday currency volatility.

6.2 Export Compliance & Automated Monthly e-FIRC Generation

For Indian exporters receiving cross-border inward remittances in USD, EUR, or GBP under RBI OPGSP/PA-CB guidelines, NovaPay automatically reconciles shipping bills and invoice metadata against AD-I partner banknostro credits. Digital Foreign Inward Remittance Certificates (e-FIRC) and Purpose Code P0802 (Software & IT Consultancy Export) compliance statements are generated automatically and uploaded to the Merchant Compliance Portal by the 5th business day of the following calendar month at zero additional cost.

6.3 International Settlement Timeline & Real-Time Sanctions Screening

Cross-border transactions settle via SWIFT gpi or local clearing rails (ACH, SEPA, Faster Payments) on a T+3 business day schedule. Every cross-border payer and beneficiary undergoes synchronous screening against OFAC SDN, UN, EU, and UK HM Treasury sanctions lists in under 200 milliseconds prior to capture.

Table 6.1: Cross-Border FX & International Compliance Matrix

FX / Cross-Border Feature	Supported Scope / Metric	Fee / SLA Timeline	Regulatory Reference
Presentment Currencies	135 Global Currencies	24-Hour Locked FX Quote	ISO-4217 Currency Codes
Settlement Currencies	14 Major Currencies (USD/INR/EUR..)	Settled to Local Bank Rail	Multi-Currency Ledger
Dynamic FX Conversion Markup	Wholesale Mid-Market Rate	+1.75% Flat FX Fee	Locked at Auth Timestamp
Cross-Border Settlement Cycle	SWIFT gpi / Local Clearing Rails	T+3 Business Days	Requires Valid Purpose Code
Sanctions Screening Latency	OFAC / UN / EU / HMT Watchlists	< 200 milliseconds (Sync)	Auto-Block on Fuzzy Match > 92%
Digital e-FIRC Issuance	Indian Export Merchants (P0802)	By 5th Business Day of Next Month	RBI PA-CB & FEMA Compliant

Policy Reference FX-603: Single cross-border invoices exceeding $10,000 USD equivalent require automated upload of the commercial invoice PDF via the Documents API (/v2/compliance/documents) prior to T+3 settlement release.

7. Automated KYB Onboarding, Transaction Tier Caps & Risk Scoring

Executive Summary: *NovaPay utilizes an AI-assisted Know Your Business (KYB) verification pipeline to onboard merchants in under 2 minutes while enforcing tiered monthly processing volume caps aligned with anti-money laundering (AML) regulations.*

7.1 120-Second Automated KYB Verification Pipeline

During merchant registration, the NovaPay KYB Engine validates corporate tax identifiers (GSTIN, Corporate PAN, CIN, or US EIN), incorporation certificates, and Ultimate Beneficial Owner (UBO) government IDs via real-time government registry APIs and OCR document forensics. Clean applications complete automated verification and receive live production API keys within 120 seconds (2 minutes). Applications flagged for manual name mismatch review are resolved by the Underwriting Desk within 4 business hours.

7.2 Monthly Transaction Volume Caps by KYC Tier

Merchant accounts are categorized into three operational KYC tiers: Tier-0 (Unverified Sandbox) accounts can only transact on test keys and have a $0.00 live processing cap. Tier-1 (Basic Digital KYC) accounts receive instant live activation and can process up to $10,000 USD (or INR 8,50,000) in gross monthly volume. Tier-2 (Full Enterprise KYB with 2 years audited financials and physical address verification) unlocks unlimited monthly processing volume, custom interchange-plus pricing, and dedicated account management.

7.3 Dormant Account Archival & Penny-Drop Re-Verification Policy

Any production merchant account exhibiting zero live transaction activity for 180 consecutive calendar days is automatically transitioned to COLD_STANDBY status. To reactivate a COLD_STANDBY account, the merchant administrator must complete fresh 2FA verification and pass an automated INR 1.00 / $0.01 bank account penny-drop name match before payouts can resume.

Table 7.1: Merchant Onboarding Tiers & Volume Thresholds

Merchant KYC Tier	Verification Requirements	Monthly Volume Cap	Activation SLA
Tier-0 (Unverified Sandbox)	Email + Mobile OTP Only	$0.00 Live (Test Mode Only)	Instant (0 Seconds)
Tier-1 (Basic Digital KYC)	Tax ID + Director ID + Penny Drop	$10,000 USD / INR 8.5L per Month	Within 120 Seconds (Auto)
Tier-2 (Full Enterprise KYB)	Audited Financials + Board Resolution	Unlimited Processing Volume	Within 24 Business Hours
Manual Underwriting Review	Triggered on OCR / Name Mismatch	Holds Live Payouts Until Cleared	Within 4 Business Hours
Dormant Account (COLD_STANDBY)	180 Consecutive Days Inactive	Payouts Paused Until Re-Verified	Instant via Penny-Drop API

Policy Reference KYC-702: When a Tier-1 merchant reaches 80% of their $10,000 monthly cap ($8,000), NovaPay triggers webhook kyc.limit_warning so the merchant can upload Tier-2 documents without checkout interruption.

8. Multi-Region Active-Active Database Topology, Circuit Breakers & DR SLAs

Executive Summary: *NovaPay's core financial ledger and AI Gateway infrastructure are engineered for zero-data-loss resilience across multi-region cloud zones, utilizing deterministic circuit breakers and sub-second automated upstream failover.*

8.1 Multi-Region Active-Active Topology & 99.995% Uptime SLA

All core payment capture, ledger, and SentinelAI gateway clusters run in an active-active configuration across AWS Mumbai (ap-south-1) and AWS Hyderabad (ap-south-2) availability zones. Ledger transactions are committed across distributed PostgreSQL/CockroachDB nodes using synchronous Raft consensus. NovaPay provides a contractual uptime SLA of 99.995% per calendar quarter (permitting less than 36 seconds of unplanned downtime per month).

8.2 Smart Circuit Breaker & Multi-Provider Failover (50ms Switch)

Both the payment switch and the SentinelAI LLM Gateway continuously monitor upstream provider health over a 30-second sliding window. If a primary upstream provider (e.g., Acquirer Switch A or Primary LLM API OpenAI) exhibits a p95 latency exceeding 2,000 milliseconds (2.0 seconds) OR an HTTP 5xx / timeout error rate exceeding 5.0%, the local circuit breaker trips to the OPEN state. Traffic is rerouted automatically within 50 milliseconds to the warm standby provider (e.g., Secondary Acquirer B or Groq Llama-3.3-70B over Server-Sent Events) without dropping in-flight client connections. Every 30 seconds, the circuit enters HALF_OPEN state and sends 3 canary probe requests to test primary recovery.

8.3 Disaster Recovery Objectives (RPO = 0s, RTO < 15s)

Because financial ledger entries and Redis write-ahead journals are replicated synchronously across regions prior to returning HTTP 200, NovaPay guarantees a Recovery Point Objective of strictly zero data loss (RPO = 0 seconds). In the event of a complete regional data center outage, BGP Anycast and Route53 health checks shift 100% of ingress traffic to the surviving region within a Recovery Time Objective (RTO) of under 15 seconds.

Table 8.1: High-Availability, Circuit Breaker & Disaster Recovery SLAs

Reliability Parameter	Target Threshold / Metric	Detection / Action Mechanism	Recovery Behavior
Contractual Uptime SLA	99.995% per Quarter	Multi-Region Active-Active	< 36s Downtime / Month
Recovery Point Objective (RPO)	0 Seconds (Zero Data Loss)	Synchronous Raft Consensus	Guaranteed Committed Writes
Recovery Time Objective (RTO)	< 15 Seconds	BGP Anycast + DNS Failover	Automated Regional Shift
Circuit Breaker Latency Trip	p95 Latency > 2,000 ms (2.0s)	30-Second Sliding Window	Trips to OPEN State
Circuit Breaker Error Trip	5xx / Timeout Rate > 5.0%	30-Second Sliding Window	Reroutes in < 50ms to Backup
Half-Open Canary Recovery	Every 30 Seconds in OPEN State	3 Sequential Probe Requests	Closes Circuit on 100% Success

Policy Reference INFRA-805: In SentinelAI Gateway deployments, when OpenAI times out (> 2,000ms), requests automatically fail over to Groq (Llama-3) while preserving Server-Sent Events (SSE) token streaming format.

9. Recurring Subscription Engine, Pre-Debit Mandates & Smart Dunning

Executive Summary: *The NovaPay Subscription & Recurring Billing Engine automates SaaS billing lifecycles, regulatory pre-debit notifications (RBI e-Mandate / UPI AutoPay), intelligent retry dunning for soft card declines, and exact per-second plan proration.*

9.1 Pre-Debit Notifications & AFA-Exempt Recurring Thresholds

To comply with central bank recurring mandate regulations, NovaPay automatically dispatches a pre-debit notification (via SMS and email) to the subscriber exactly 24 hours prior to executing any scheduled recurring charge. Recurring transactions up to INR 15,000 (or $200 USD) per charge execute seamlessly without Additional Factor Authentication (AFA). Any recurring invoice exceeding INR 15,000 automatically generates an interactive 3DS2 / OTP approval link sent to the customer.

9.2 Smart Dunning Retry Schedule for Soft Declines

When a scheduled subscription charge fails due to a temporary soft decline (e.g., insufficient funds, issuer switch timeout, or daily card limit exceeded), the Smart Dunning Engine holds the subscription in a GRACE_PERIOD state and automatically retries the charge up to 4 times on a deterministic schedule: Retry 1 on Day +1, Retry 2 on Day +3, Retry 3 on Day +5, and Final Retry 4 on Day +7 at optimal issuer-approval daytime hours (11:30 AM local time). If all 4 retries fail, the subscription status transitions to PAST_DUE and emits webhook subscription.past_due.

9.3 Exact Per-Second Mid-Cycle Upgrade Proration

When a customer upgrades or downgrades their subscription tier mid-billing cycle, the billing engine calculates proration credits down to the exact second using the formula: credit = (unused_seconds / total_cycle_seconds) * current_plan_price. The net delta is invoiced immediately upon upgrade or applied as an immutable ledger credit toward the next renewal cycle upon downgrade.

Table 9.1: Subscription Mandate & Smart Dunning Schedule

Subscription Workflow	Rule / Schedule	Threshold / Formula	Resulting State / Webhook
Pre-Debit Customer Notice	24 Hours Before Charge	Mandatory for All Mandates	Webhook: mandate.notified
AFA-Exempt Auto-Debit Cap	Max INR 15,000 / $200 USD	Above Cap Requires Customer 2FA	Auto-Executes on Due Date
Smart Dunning Retry 1 & 2	Day +1 and Day +3 (11:30 AM)	Triggered on Soft Decline Only	Status: GRACE_PERIOD
Smart Dunning Retry 3 & 4	Day +5 and Day +7 (11:30 AM)	Final Automated Recovery Attempts	Transitions to PAST_DUE if Failed
Hard Decline (Stolen/Closed Card)	Zero Retries (Immediate Stop)	Reason Codes: 41, 43, 05-Hard	Webhook: subscription.halted
Mid-Cycle Plan Proration	Exact Per-Second Granularity	(unused_sec / total_sec) * price	Immediate Delta Invoice / Credit

Policy Reference SUB-904: Hard declines (lost/stolen card or revoked mandate) bypass Smart Dunning immediately to avoid card network excessive-retry penalties ($0.10 per invalid retry after hard decline).

10. Enterprise Pricing Matrix, Incident Response SLAs & Service Credits

Executive Summary: *This section outlines NovaPay's transparent transaction pricing structure, 24/7/365 engineering incident response SLAs, and contractual financial service credits awarded in the event of an SLA breach.*

10.1 Transparent Platform Transaction Pricing

Standard domestic transactions processed via credit cards, debit cards, netbanking, and wallets are billed at a flat platform fee of 2.00% per successful capture (plus applicable statutory taxes). Standard UPI checkout transactions carry a 0.00% platform MDR fee. International credit cards and American Express cards are billed at 3.00% per transaction (plus the 1.75% FX conversion fee if currency settlement differs). There are zero setup fees, zero minimum monthly commitments, and zero annual maintenance charges on Standard and Growth plans.

10.2 24/7 Engineering Incident Response SLAs by Severity

NovaPay Technical Support classifies production incidents into three severity levels: Severity-1 (Critical Production Outage or >10% payment capture drop) guarantees a dedicated SRE engineering response within 15 minutes 24/7/365 via Slack Connect and PagerDuty bridge. Severity-2 (Partial Degradation, webhook delays, or dashboard unavailability) guarantees engineering response within 1 hour. Severity-3 (General technical integration queries, sandbox issues, or reconciliation questions) guarantees response within 8 business hours.

10.3 Contractual SLA Breach Service Credits

If NovaPay fails to meet its 99.995% quarterly uptime commitment for Enterprise merchants, affected merchants are eligible for automatic fee service credits applied against their next monthly invoice: (a) Quarterly uptime between 99.90% and 99.994% awards a 10% service credit on monthly platform fees; (b) Quarterly uptime between 99.50% and 99.89% awards a 25% service credit; and (c) Quarterly uptime below 99.50% awards a 50% service credit.

Table 10.1: Pricing Schedule, Support Response SLAs & Fee Service Credits

Category / Tier	Scope / Condition	Rate / Response SLA	Notes & Credit Terms
Domestic Cards & Netbanking	Visa, Mastercard, RuPay, 50+ Banks	2.00% per Captured Txn	Zero Setup or Annual AMC Fee
Standard UPI Checkout	UPI Intent & Collect Flows	0.00% Platform Fee	Instant T+1 Settlement Included
International Cards & Amex	135 Currencies Supported	3.00% + 1.75% FX Markup	Includes 200ms Sanctions Check
Severity-1 (Critical Outage)	Core Capture Down / >10% Error	Within 15 Minutes (24/7/365)	Direct SRE Slack + PagerDuty Bridge
Severity-2 (Partial Degradation)	Webhook Delay / Dashboard Slow	Within 1 Hour (24/7)	Priority Engineering Escalation
Severity-3 (General Query)	SDK / Sandbox / Billing Docs	Within 8 Business Hours	Standard Merchant Support Portal
SLA Breach Credit Tiers	<99.995% / <99.90% / <99.50%	10% / 25% / 50% Fee Credit	Auto-Applied to Next Monthly Invoice

Policy Reference SLA-1001: To claim an SLA credit or escalate a Sev-1 ticket programmatically, call POST /v2/support/incidents with header X-Priority: SEV1 or email sre-oncall@novapay.enterprise.